Rebuttal for CoRL Submission #2468

We thank the reviewers and Area Chair for their constructive questions. The additional analyses strengthen the paper's central conclusion that an online, tactile-supervised future-contact prior provides information beyond nominal task progress and helps a diffusion policy overcome pre-contact stall. They also identify the mechanisms responsible for this gain, motivate two cleaner implementation refinements—fixed bounded loss weighting and tactile-only masking—and establish a stricter fixed-mount evaluation protocol.

TABLE I
ADDITIONAL OPERATOR-HELD SWIPE RESULTS. SUCCESS IS ONE COMPLETE ROLLER ROTATION WITHIN 60 S. RESULTS ARE KEPT SEPARATE FROM FIXED-MOUNT TRIALS; WILSON 95% INTERVALS ARE SHOWN.

Policy or isolated change	Success/N	95% CI
Legacy Full	9/10	[.596, .982]
Phase-DP†	12/20	[.387, .781]
Progress-DP†	14/20	[.481, .855]
Learned weight $\rightarrow$ fixed $1+c$	15/25	[.407, .766]
Whole-lowdim $\rightarrow$ tactile-only mask	13/20	[.433, .819]
No-gating retrain	8/20	[.219, .613]
Unbounded $\rightarrow$ bounded gate	7/20	[.181, .567]

Q1: Contact anticipation versus task progress. We thank the reviewer for pointing us to Hierarchical Diffusion Policy (Wang et al., T-RO 2025) and CompILE (Kipf et al., ICML 2019). These relevant works were not cited in the submitted version; we will add them to Related Work and clarify that hierarchical phase decomposition and predictive contact conditioning provide complementary inductive biases. Phase and progress are useful, inexpensive conditions: they encode the nominal task order without requiring tactile labels or an additional predictor. However, they are time-indexed rather than state-conditioned. The same phase or progress value is produced even if execution stalls, contact geometry changes, or the object is displaced. VPCE trades this simplicity for physical grounding: it predicts a continuous 15-step future-contact horizon online from current wrist RGB and recent robot motion, supervised by tactile-derived contact events. Unlike a hierarchical policy, VPCE neither switches sub-policies nor requires a discrete high-level controller; it continuously conditions the same diffusion action generator on whether contact is becoming imminent. Because the archived real-time fallback for Phase/Progress advanced its private clock faster than physical control time, we treat these policy-level rows as preliminary diagnostics rather than final claims; the main conclusion is instead supported by the predictor diagnostic in Table II and the fixed-mount study now being completed.

As a predictor-level diagnostic on held-out trajectories, however, VPCE is not explained by progress alone: Progress captures some timing, but vision, motion, and tactile-derived contact supervision add information beyond normalized trajectory time. This distinction also explains the synergy with tactile feedback. VPCE supplies information before touch becomes informative—*when* to commit to contact—whereas tactile observations regulate *how* to execute after contact. The submitted 2×2 comparison separates these roles: on SWIPE, the contact-prior-only policy reaches 85% versus 10% for tactile-only DP, and the full method reaches 90%; on WIPE/INSERT, the full method reaches $70/45\%$, compared with $50/30\%$ for prior-only and $60/15\%$ for tactile-only policies. Thus contact anticipation and tactile feedback are complementary rather than competing representations. The zero-shot results further delineate the current transfer boundary: source-teacher AUC/F1 is $.500/.095$ on board and $.662/.666$ on card, while target adaptation raises board performance to $.897/.750$. The current target-domain teacher uses 60 demonstrations with tactile recordings per downstream task; contact targets are generated automatically from the recorded tactile signals, without manual frame-level annotation. The current evidence therefore supports a contact-specific signal beyond progress, with target adaptation improving transfer across new visual/contact domains.

TABLE II
CONTACT PREDICTION AUC/F1 USING PHASE, PROGRESS, OR VPCE.

Task	Phase-4	Progress-20	VPCE
SWIPE	$.670/.000$	$.853/.597$	$\mathbf{.963/.759}$
WIPE	$.666/.000$	$.743/.406$	$\mathbf{.885/.705}$
INSERT	$.684/.000$	$.808/.539$	$\mathbf{.946/.735}$

Q2: Loss-weight optimization. We thank the reviewer for identifying this optimization ambiguity. The original design intuition was to emphasize sparse but control-critical approach-to-contact transitions. With uniform denoising loss, abundant and relatively easy free-space samples can dominate optimization, even though small action errors near contact determine whether engagement succeeds. We therefore used the VPCE mean probability c to increase the gradient contribution of contact-imminent samples. The learnable γ was intended to adapt the global strength of this emphasis across tasks, while the residual δ was intended to compensate for the fact that contact probability and action difficulty are not identical; $\mathrm{softplus}(\gamma)$ kept the base gain non-negative. We agree, however, that optimizing these parameters through the same weighted objective creates a downward incentive. Checkpoint inspection confirms that the resulting weight is approximately 1.000–1.005: the lower bound prevents harmful down-weighting, but largely reduces the term to ordinary DP loss. We retain the original emphasis intuition while removing this ambiguity with the fixed bounded schedule $w(c) = \mathrm{clip}(1 + c, 1, 2)$, which smoothly maps free-space samples to weight 1 and high-contact-probability samples to at most 2, without a learned collapse direction. The updated $15/25$

result confirms that this well-posed refinement preserves real-robot functionality. We further evaluated the fixed family $w_\alpha(c) = \mathrm{clip}(1 + \alpha c, 1, 1 + \alpha)$ in a matched SWIPE sweep with 10 real-robot rollouts per value. Success was $10/10$ for α=1, compared with $7/10$, $6/10$, and $3/10$ for α=0.5, 0.25, and 2, respectively. Thus moderate contact emphasis works best in this setting: weaker schedules provide less rebalancing, whereas α=2 over-emphasizes contact-imminent samples. We select α=1 and will report this sensitivity study in the revision. Together with the submitted no-loss-weighting result (8/10), these experiments show that the central contact-anticipation effect does not depend on the original learned weighting term, while the fixed schedule retains the intended emphasis on contact-critical samples without an optimization shortcut.

Q3: Masking and free-space motion. We thank the reviewer for highlighting that Eq. (9) conflated distinct low-dimensional signals. The original post-encoder block included proprioception, gripper width, and tactile features, although the intended inductive bias was only to suppress uninformative pre-contact touch. We therefore ran a controlled mask-scope retraining that changes only this routing: the revision always retains proprioception and gripper width and masks only tactile/contact features. It obtains $13/20$ on SWIPE. Qualitatively, preserving proprioception produces more coherent free-space approach motion and avoids the endpoint-seeking/repositioning behavior observed with the whole-lowdim mask, while contact-adaptive tactile masking still suppresses pre-contact sensor variation. Action-chunk prediction and receding-horizon execution provide additional temporal coherence. This result supports masking as tactile routing, rather than as suppression of robot state.

Q4: Gating and component ablations. We thank the Area Chair and Reviewer Fcwd for emphasizing that the component ablations should be visible in the main paper rather than only in the appendix; this will make the evidence for each design choice easier to assess. The submitted ablation showed that removing horizon conditioning or masking reduced SWIPE to $2/10$ and $0/10$, whereas removing gating gave $9/10$ and removing loss weighting gave $8/10$. Together with the new no-gating $(8/20)$ and bounded-gate $(7/20)$ retrains, these experiments identify a clear division of labor: future-contact horizon conditioning and contact-adaptive masking are the main drivers of task success; loss weighting is a secondary training emphasis; and gating is an auxiliary, task-dependent modulation rather than a required source of the contact-anticipation gain. We will move the compact component-ablation table and this interpretation into the main text in the revision, and retain the full behavioral analysis in the appendix.

Q5: Operator-held evaluation protocol. We thank the reviewer for highlighting that the supplementary video did not make this protocol and its rationale sufficiently clear. Holding the reader and foam was an intentional safety measure: mechanical compliance reduces the risk of damaging the fingertip tactile sensors, task props, or robot during failed high-force contacts. The same held-object protocol, success criterion, and 60-s timeout were applied across methods. We increased the number of trials and report success counts with Wilson intervals to reduce and expose the remaining sampling uncertainty. Under this shared protocol, the visual and naive visuo-tactile baselines still exhibited pre-contact stall, whereas the VPCE-conditioned policies committed to engagement; thus the comparison directly tests the paper's central question of whether anticipation helps initiate contact. We nevertheless agree that human compliance may make alignment easier and should be isolated rather than treated as part of the policy. We are therefore completing a separate rigid-mount comparison of DP-VT, Legacy Full, and fixed-loss VPCE-DP with identical initialization, controller, timeout, and randomized N=20 trials, and will report it separately rather than pooling the two protocols. We will clarify this limitation and investigate compliant, force-limited fixtures as a safer standardized evaluation setup in future work.

Q6: Contact-label calibration and contact direction. We thank the reviewer for raising this important extension. In the current setup, two planar fingertip tactile sensors and the gripper/task geometry yield a repeatable dominant interaction direction. We therefore derive the binary contact event from the temporal change of a base-frame z-force proxy; the same threshold of 0.095 is reused for most tasks. For the few domains with a different sensor scale, the threshold is selected once from the training-demonstration force-change distribution and then held fixed. All frame labels are generated automatically from the tactile stream; no manual frame-level contact annotation is required. This is sufficient for the present tasks because VPCE predicts whether contact is imminent, rather than regressing the full contact wrench. We agree that dexterous hands, curved tactile surfaces, and contacts with changing normals require direction-agnostic or local-frame labels. A direct extension is to threshold the resultant change $\|\Delta \mathbf{F}\|_2$, or combine local normal and shear changes with hysteresis or learned change-point detection. Importantly, this replaces only the tactile-to-contact label adapter: the VPCE predictor, future-contact horizon, and policy-conditioning framework remain unchanged. We will add a threshold sensitivity analysis and investigate calibration-light multi-axis labeling in the revision and future work.

Scope. Target adaptation defines the current transfer boundary. The three tasks provide complementary evidence across precise engagement, sustained surface contact, and insertion. SWIPE shows the largest gain, while WIPE and INSERT test whether the same prior remains useful when post-contact regulation becomes increasingly important. We report Wilson intervals and avoid extrapolating beyond the evaluated task set.